

Supplementary Figures Legends

Suppl. Figure 1. A) WB analysis of 9464D and 975A2 protein extracts after *Atg7* downregulation using two different shRNA (shAtg7#1, shAtg7#2) or with control (shCtrl) by lentiviral infection. Representative immunoblotting of Atg7, p62 and LC3 is shown. Actin was used as a loading control. Densitometric analysis of Atg7, LC3II and p62 expression levels over Actin is shown. Data $\pm$ SEM are presented, and significance is calculated using one-way ANOVA (* $p < 0.1$, ** $p < 0.01$; $n=3$). **B)** Representative immunoblotting of Ulk1, pAtg14, Atg14, and LC3 in protein extracts of 9464D and 975A2 cell lines treated with the indicated concentrations of SBI-0206965 and CQ for 48 hours respectively. Densitometric analysis of p-Atg14 over Atg14/Actin is shown. Data $\pm$ SEM are presented, and significance is calculated using one-way ANOVA (** $p < 0.01$; $n=3$). **C)** Analysis of 3D growth of 9464D cells after *Atg7* downregulation by lentiviral infection (shAtg7#1 or shCtrl). The spheroid area was calculated at the indicated time points. Data are presented as mean $\pm$ SEM calculated over 24 hours (ns, not significant; two-way ANOVA, $n=6$). Representative images are shown. **D)** Representative images of 9464D spheroids treated with 40 μ M CQ and acquired at the indicated time points. **E)** Analysis of cell viability of both 9464D and 975A2 NB cell lines treated with the indicated doses of SBI-0206965 or CQ 24 hours after plating. Cells were analyzed by MTS assay at the indicated time points. Bar graphs show the 490 nm absorbance mean $\pm$ SEM calculated over control cells (* $p < 0.1$, ** $p < 0.01$; two-way ANOVA, $n=3$). **F)** Analysis of cell viability of SH-SY5Y, LAN-5, and IMR32 cells treated with the indicated doses of SBI-0206965 by MTS assay as in E. Data are presented as the mean $\pm$ SEM calculated over control cells (* $p < 0.1$, ** $p < 0.01$; two-way ANOVA $n=3$); **G)** Analysis of spheroid area of SH-SY5Y, LAN-5 and IMR32 cells after treatment with different concentration of SBI-0206965. The spheroid area was calculated at the indicated time points. Data are presented as mean $\pm$ SEM calculated over 24 hours (two-way ANOVA, $n=3$).

Supp. Figure 2. A) Representative immunoblotting of pAtg14 and Atg14 in protein extracts of 9464D cell line treated with the indicated concentrations of ULK-101 for 24 hours. Densitometric analysis of p-Atg14 over Atg14/Hsp90 is shown. Data $\pm$ SEM are presented, and significance is

calculated using one-way ANOVA (**p< 0.01; n=3). **B)** MHC-I expression was determined by flow cytometry after treatment with the indicated doses ULK-101 for 24 and 48 hours in the 9464D cell line. Data $\pm$ SEM are presented as mean fluorescence intensity (MFI) normalized over shCtrl cells (one-way ANOVA followed by Tukey post hoc test). (n=3).

Suppl. Figure 3. A) Volcano plot of mass-spectrometry identified proteins derived from ex vivo tumor tissue. Fold Change of shAtg7 versus shCtrl modulated proteins are represented (Red: upregulated, Blu: downregulated DEPs in shAtg7 tumors). **B)** Network analysis of shAtg7 versus shCtrl-derived tumors. Red: upregulated Blu: downregulated proteins, Size: -log p-value. **C)** Percentage of CD69⁺ PD1⁺ and CD69⁻ PD1⁺ cells of both CD4⁺ and CD8⁺ T cell populations collected from 9464D bearing C57BL/6 mice treated with CQ or vehicle respectively. **D-E)** Immunophenotype of infiltrating T cells (CD45⁺, CD3⁺, CD4⁺, CD8⁺ CD44⁺/CD62L⁻ effector memory (EM)) in shAtg7#2 and shCtrl 975A2 tumors evaluated by flow cytometry. The number of events was normalized on tumor weight. Data are presented as mean $\pm$ SEM calculated by Mann Whitney test. (n>10) *p< 0.05.

Supp. Figure 4. A) CD8 expression levels of sorted T cells (n=4). Representative dot plot for CD8 expression of CD3⁺ population. **B)** Functional activity in long-term killing assays of PRAME-TCR-T cells against THP-1 and HL-60 cells in the E:T ratio of 1:2 (n=3).

Supp. Figure 5. A) Schematic representation of T-cells transduced with a third-generation CAR directed against GD2⁺ expressed on NB cells. **B)** GD2.CAR expression levels on non-transduced (NT) control and GD2.CAR T cells (n=3). Representative dot plot of NT control cells and GD2.CAR transduced T cells. Cells were surface stained with 1A7 antibody (anti-GD2.CAR) with subsequent staining of PE-labeled secondary antibody. **C)** Functional activity in long-term killing assays of GD2.CAR T cells against SH-SY5Y in an E:T ratio of 1:2 (n=3). **D)** Cytotoxic effects of 5 μ M SBI-0206965 (SBI) and 40 μ M CQ treatment in CHLA-255-GFP⁺ and SH-SY5Y-GFP⁺ spheroids respectively. Data are presented as mean $\pm$ SEM calculated by two-way ANOVA (n=3 experiments). **(E-F)** Long-term killing assays of NT cells co-cultured at different E:T ratios

with 24 hours SBI-0206965 (5 μ M)-pretreated spheroids of CHLA-255-GFP⁺ and SH-SY5Y-GFP⁺ cells. Data are presented as mean $\pm$ SEM calculated by two-way ANOVA (n=3 experiments). **G)** Long-term killing assays of GD2.CAR T cells co-cultured at different E:T ratios with 24 hours SBI-0206965 (5 μ M)-pretreated spheroids of CHLA-255-GFP⁺ and SH-SY5Y-GFP⁺ cells respectively. Data are presented as mean $\pm$ SEM calculated by two-way ANOVA (n=3 experiments). *p< 0.05, **p< 0.01. **H)** Representative images and average (AVG) GFP intensity of 24 hours CQ-pretreated SH-SY5Y-GFP⁺ spheroids co-cultured with GD2.CAR T cells at 1:3 E:T ratios and for the indicated time points. Data are presented as mean $\pm$ SEM calculated by two-way ANOVA (n=3 experiments). *p< 0.05, **p< 0.01. **I)** Average (AVG) GFP intensity of 24 hours CQ-pretreated CHLA-255-GFP⁺ spheroids co-cultured with GD2.CAR T cells at 1:3 E:T ratios and for the indicated time points. Data are presented as mean $\pm$ SEM calculated by two-way ANOVA (n=3 experiments). *p< 0.05. **J)** Long-term killing assays of GD2.CAR T cells co-cultured at 1:3 and 1:2 of E:T ratios with shAtg7 vs shCTRL SH-SY5Y-GFP⁺ spheroids. Data are presented as mean $\pm$ SEM calculated by two-way ANOVA (n=3 experiments). *p< 0.05, **p< 0.01, ***p< 0.001.

Supp. Figure 6. A) Tumor growth analyses in mice receiving NT cells +PBS or combined with CQ. Statistical analyses by two-way ANOVA followed by Šidák's multiple comparisons test. **B)** Flow cytometry analyses of tumor infiltrated cells (GFP⁺) after spleen dissociation of mice receiving GD2.CAR T or NT cells and CQ or PBS respectively. Data are presented as mean $\pm$ SEM. Statistical analysis by one-way ANOVA. **(C)** Representative gate strategy of CD4⁺, CD8⁺ (both gated on CD3⁺ cells), CD95⁺ and HLA-DR⁺ T cells (both gated on CD4⁺ or CD8⁺ cells) in CD45⁺/FITC⁻ spleen infiltrating cells. **(D)** Representative gate strategy of GD2.CAR T CD4⁺, CD8⁺ (both gated on CD3⁺ cells), and HLA-DR⁺ T cells (both gated on CD4⁺ or CD8⁺ cells) in CD45⁺/FITC⁻ spleen infiltrating cells. **E-F)** Percentage of CAR T cells, CD4⁺, CD8⁺, and HLA-DR⁺ cells gated on GD2.CAR T cells isolated from spleens of PBS or CQ-receiving mice. Spleens were collected at the final time point and flow cytometry analyses were performed. Statistical analysis by unpaired Student's t-test. *p< 0.05, **p< 0.01. **G-H-I)** Frequency of positive CD4 and CD8 T cells, frequency of positive KI67 and HLA-DR in both CD4⁺ and CD8⁺ cells, frequency of positive PD1, Lag3 or TIM3 in both CD4 and CD8 T cells. Statistical analysis

by unpaired Student's t-test. **J)** Frequency of positive PD1, Lag3 and TIM3 in both CD4⁺ and CD8⁺ T cells. Statistical analysis by unpaired Student's t-test.